

Predicting Nifty50 Directions via Headline Sentiment Mining

Summary

Domain: Financial Text Mining & Quantitative Signals

Target Asset: Nifty 50 Index (NSE)

1. Key Finding from Data Exploration

Analyzing 3,348 headline records aggregated over 132 trading sessions revealed a clear directional pattern: **aggregated keyphrase sentiment closely aligns with same-day index performance.**

- **Up Days:** On days when the Nifty 50 closed higher than its opening price ($\text{Close} > \text{Open}$), the average daily sentiment score was **+21.63**.
- **Down Days:** On days when the index closed lower ($\text{Close} < \text{Open}$), the average sentiment dropped to **+14.81**.
- **Sentiment Spread:** The resulting +6.82 point difference indicates that domain-specific keyphrase scoring effectively captures same-day market momentum.

2. Part A: Same-Day Directional Accuracy vs. Baseline

To evaluate same-day directional prediction ($\text{Target}_t = 1$ if $\text{Close}_t > \text{Open}_t$), the rule-based dictionary approach was benchmarked against the majority-class baseline:

- **Majority-Class Baseline Accuracy: 53.85%** (achieved by predicting Up for every trading day).
- **Part A Rule-Based Sentiment Accuracy: 59.85%**.
- **Performance Gain:** Using a targeted dictionary of 200+ financial keyphrases produced a **+6.00% absolute accuracy boost** over the baseline.

3. Part B: Best Next-Day Model & Accuracy vs. Baseline

To predict next-day price direction ($\text{Target}_{t+1} = 1$ if $\text{Close}_{t+1} > \text{Open}_{t+1}$), machine learning classifiers were trained using rolling sentiment exponential moving averages (`sent_ema3`, `sent_ema7`), lagged price movements, and market volatility indicators via time-series cross-validation (`TimeSeriesSplit`):

- **Next-Day Majority-Class Baseline Accuracy: 53.85%**.
- **Best Part B Model (Logistic Regression): 50.00%** test accuracy.
- **Alternative Model (Tuned Random Forest): 30.77%** test accuracy.
- **Key Takeaway:** While simple linear classification outperformed complex tree architectures (50.00% vs. 30.77%), neither surpassed the baseline. This demonstrates how tree-based algorithms overfit on limited time-series horizons ($N = 132$ days).

4. What I Would Do Differently with More Time

Given additional development time, the primary objective would be scaling the dataset from 132 trading days to a multi-year news corpus ($N > 5,000$ days) to prevent model overfitting. Furthermore, instead of relying solely on daily static aggregations, I would implement precise intra-day timestamp alignment—mapping hourly headline sentiment directly to subsequent 15-minute and 30-minute market candle movements. Finally, moving beyond dictionary string matching to implement true financial NLP architectures (such as FinBERT embeddings) would allow us to capture deeper contextual sentiment.